University *of* Maryland School of Medicine

"Wait, when was QC last run???" Evaluating MFI drift after morning QC and its impact on unmixing.

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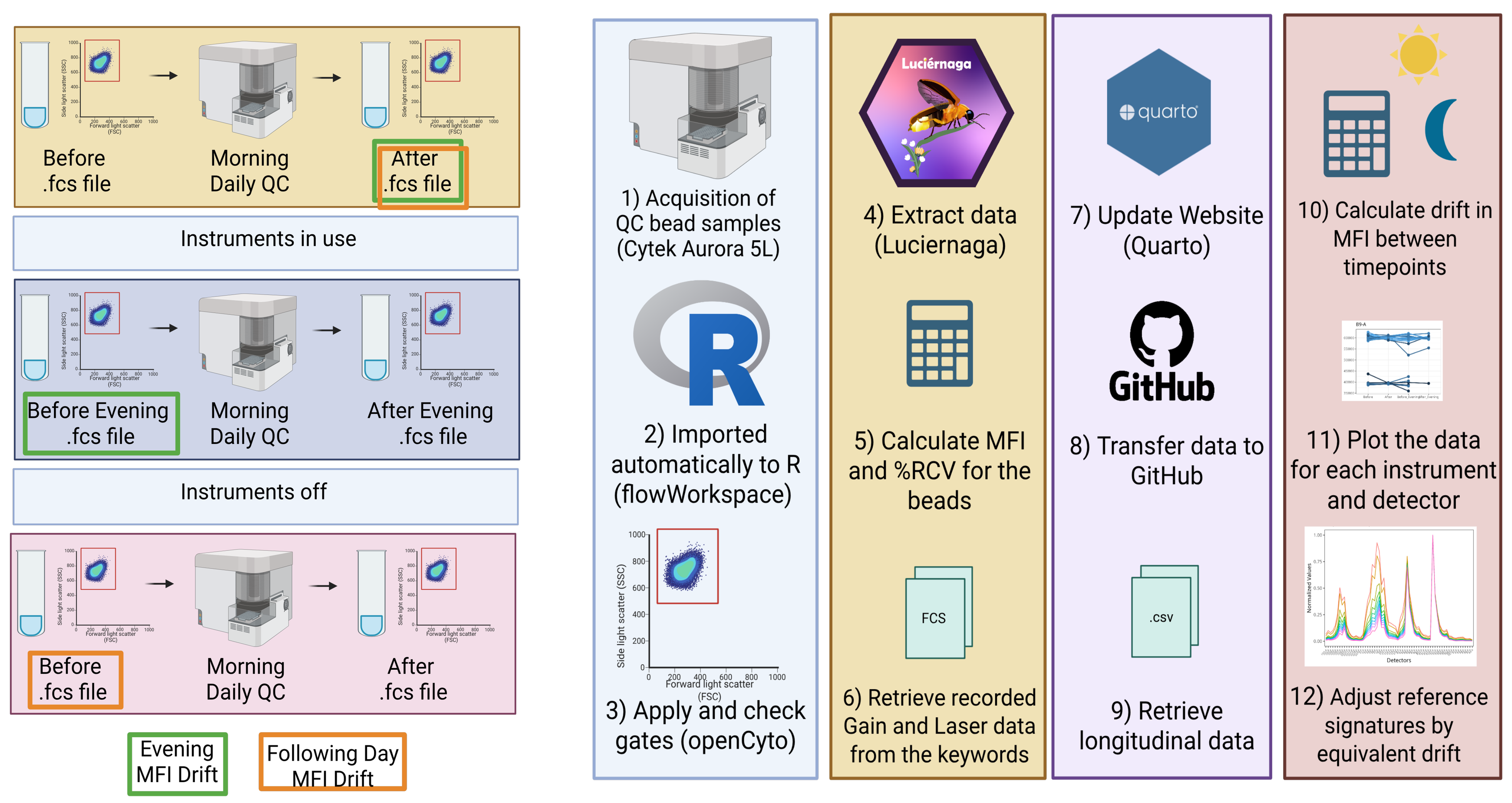
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Introduction

At our core, quality control (QC) beads are run daily on Cytek Aurora instruments as part of the instrument startup process. Based on the beads initial observed MFI values for each detector, the SpectroFlo software adjusts the gains and laser settings of the instrument to ensure that the after-QC MFI values match lot-specific thresholds. This accounts for instrumental changes, allowing specimens acquired on different days to be comparable, and reducing the frequency of MFI-based batch effects in large spectral flow cytometry panels. We recently implemented a website to monitor daily QC for our instruments at our core (https://umgccfcss.github.io/InstrumentQC/), visualizing longitudinal changes in gain, RCV, and bead MFI values before and after daily QC. In the process, we observed various detectors for which MFI values pre-QC were consistently different from the MFI values observed after QC the day before. While MFI was reset to baseline at the following QC, we were curious whether these observed MFI drifts would have already occurred by the evening before, or were a result of the instrument shutdown. Additionally, we wanted to evaluate whether these drifts were sufficient to impact unmixing when samples were acquired in the evening compared to shortly after morning QC.

Methods


Before .fcs file
Morning Daily QC
After .fcs file
Instruments in use
Before Evening .fcs file
Morning Daily QC
After Evening .fcs file
Instruments off
Before .fcs file
Morning Daily QC
After .fcs file
Evening MFI Drift
Following Day MFI Drift
1) Acquisition of QC bead samples (Cytek Aurora 5L)
2) Imported automatically to R (flowWorkspace)
3) Apply and check gates (openCyto)
4) Extract data (Luciernaga)
5) Calculate MFI and %RCV for the beads
6) Retrieve recorded Gain and Laser data from the keywords
7) Update Website (Quarto)
8) Transfer data to GitHub
9) Retrieve longitudinal data
10) Calculate drift in MFI between timepoints
11) Plot the data for each instrument and detector
12) Adjust reference signatures by equivalent drift

Results

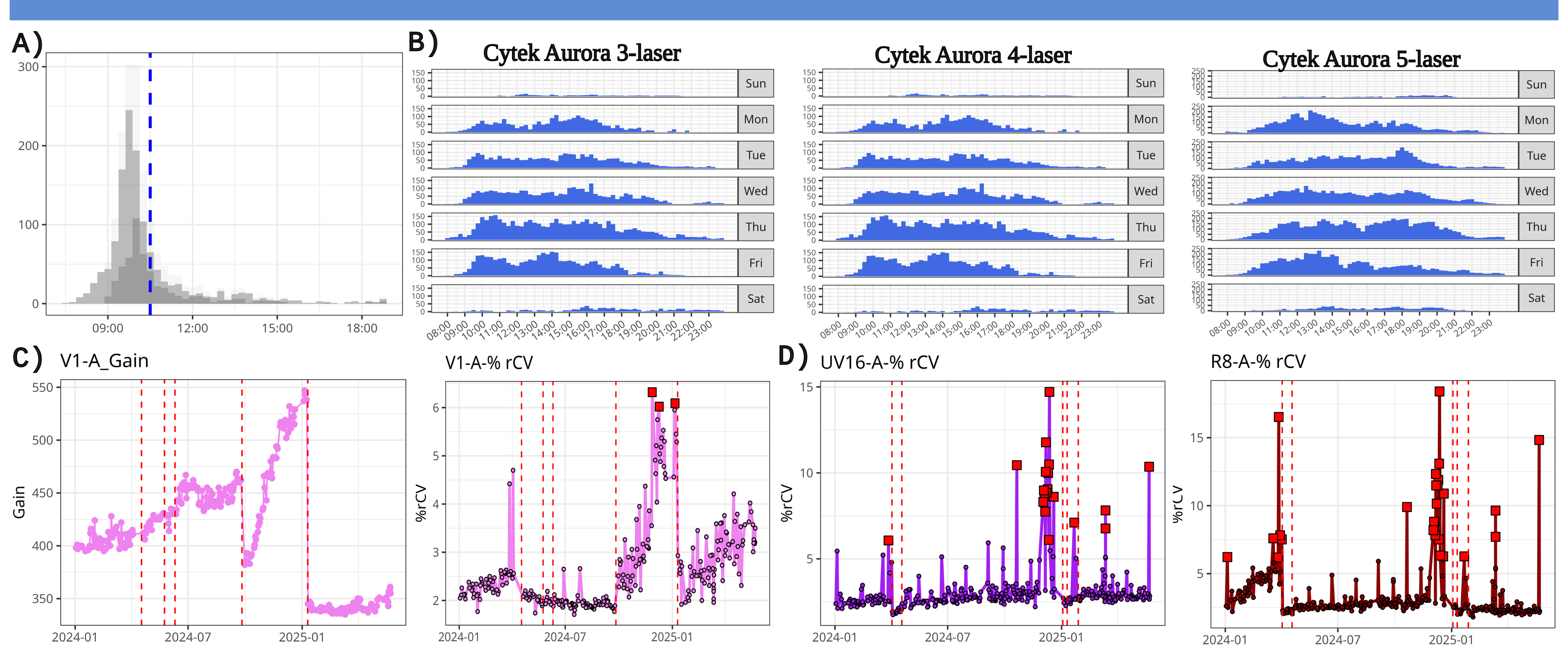

A)
B)
Cytek Aurora 3-laser
Cytek Aurora 4-laser
Cytek Aurora 5-laser
C) V1-A_Gain
V1-A-% rCV
D) UV16-A-% rCV
R8-A-% rCV

Figure 1. Automated QC processing, tracking and visualization allows for long-term monitoring of instrument performance. **A)** The histogram depicts when daily QC was performed, as recorded in the .fcs file metadata. **B)** The histogram plots depict the typical user hours shown by hour and weekday for each instrument, based on total recorded instrumental sit flushes in the Application Log. **C)** The Levey-Jennings plots depicts change in gain and %RCV representative of a failing laser, with field service engineer visits appearing as red dashed lines. **D)** The Levey-Jennings shows spikes in %RCV associated with QC beads exposure to bleach.

Results

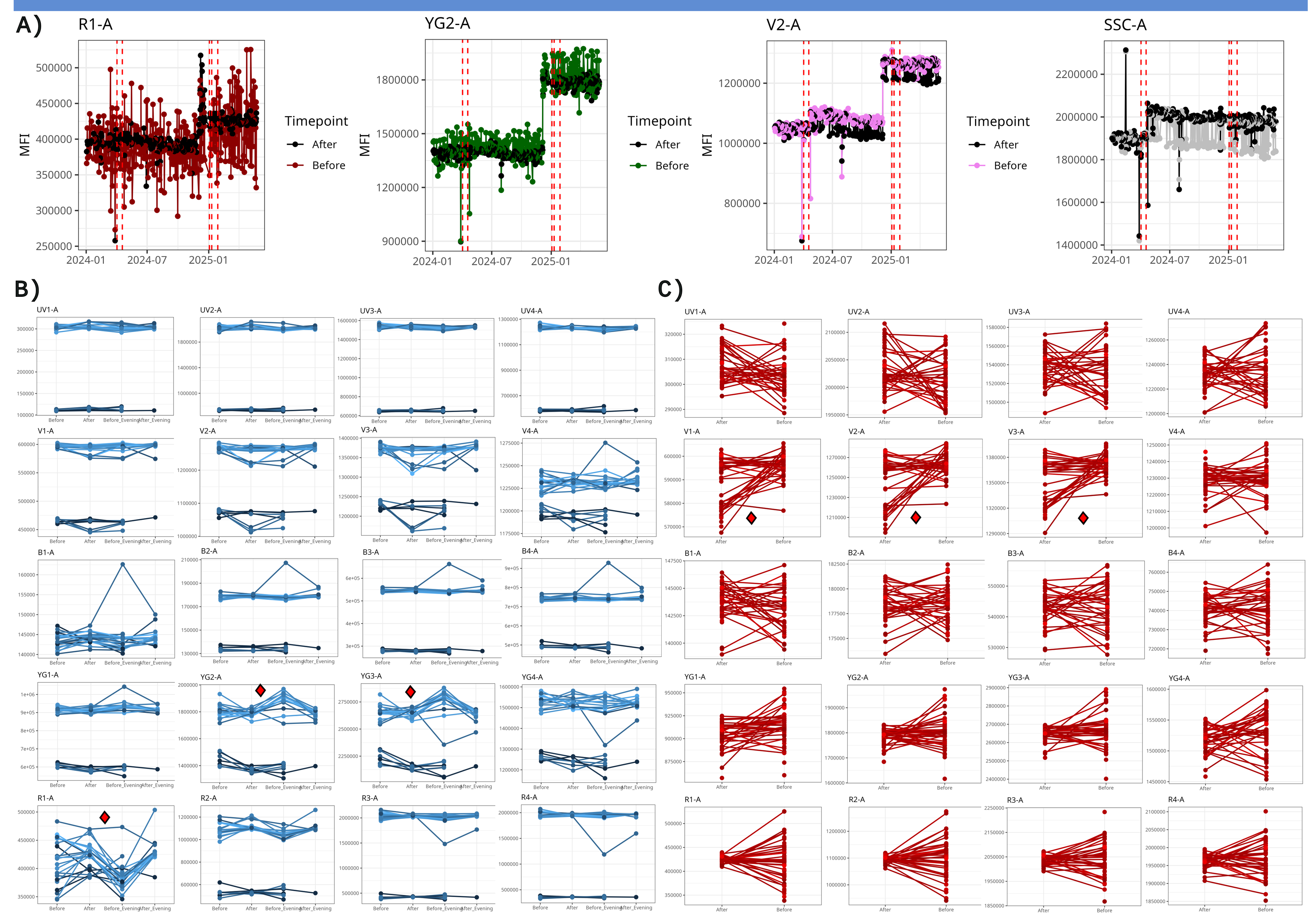

A)
R1-A
YG2-A
V2-A
SSC-A
Timepoint
After
Before
B)
C)

Figure 2. Drift in MFI can be observed by evening, but are more pronounced the folowing day. A) Typical MFI drift pattern observed for the Cytek Aurora 5-laser. **B)** The lineplots depict the MFI measurements of QC beads acquired before and after both morning and evening QC. **C)** The lineplots depict the recorded MFI measurements of QC beads after morning QC, and before morning QC the following day.

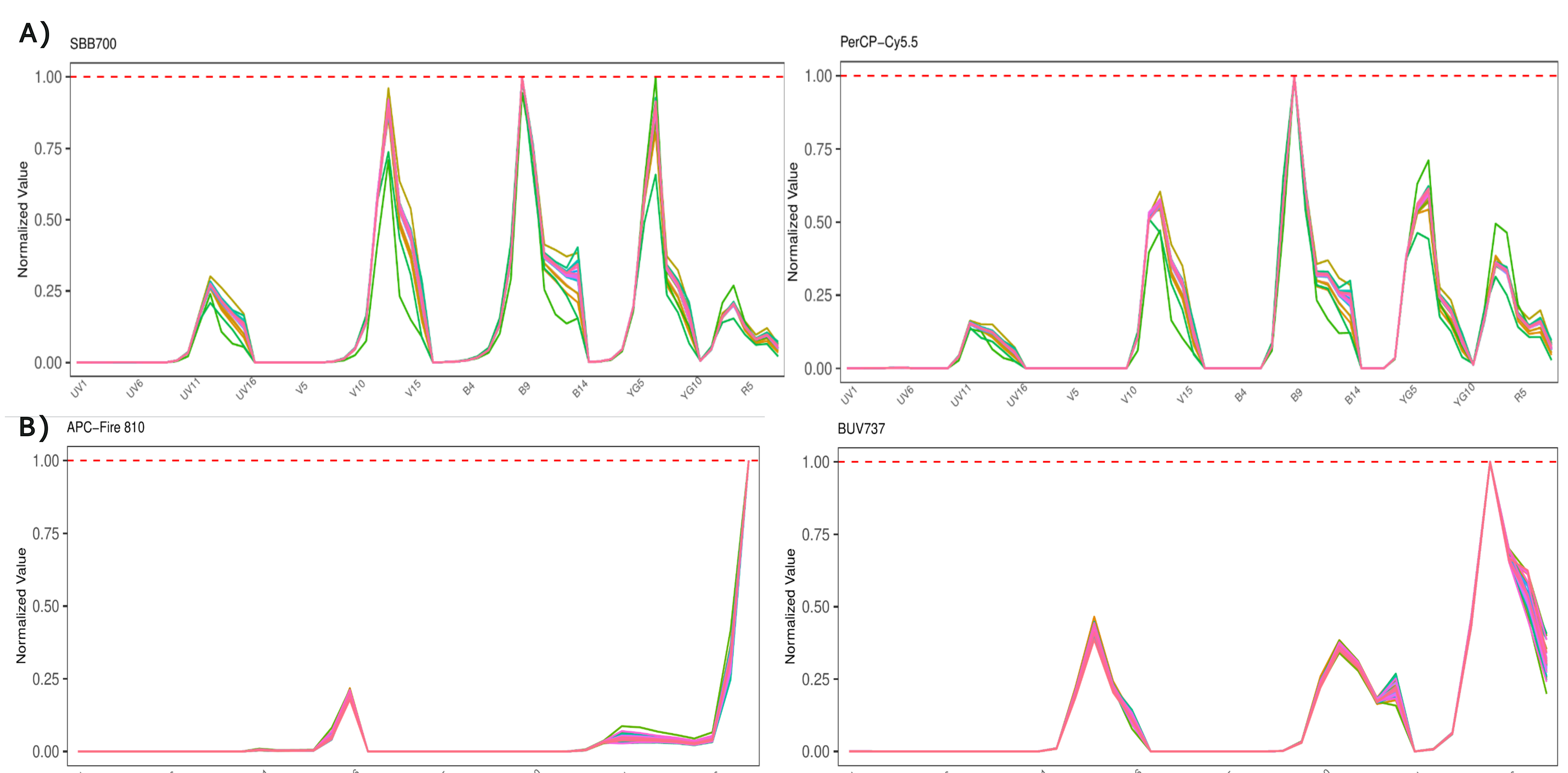

A) SBB700
PerCP−Cy5.5
B) APC−Fire 810
BUV737
Normalized Value

Figure 3. Simulated shifts in reference signatures arising from observed MFI drifts. Evening and morning after MFI drifts were calculated for each instrument for a six-month period, and fluorophore reference signatures were adjusted to acccount for the observed drift in detector. Line plots with each day a different color. **A)** shows two examples from the 5L instrument were variation was observed, while **B)** shows more typical minimal changes

Acknowledgements

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Results

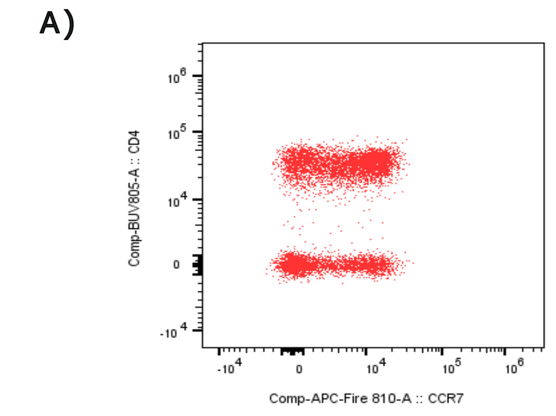

A)
Comp-BUV805-A :: CD4
Comp-APC-Fire 810-A :: CCR7

Figure 4. Simulated observed changes applied to full-stained samples and reference controls did not alter unmixing of a 29 color panel. A) Shows the original SpectroFlo unmixing on the experiment day. **B)** Shows the unmixing in SpectroFlo after shifting the unmixing controls MFI by the observed drift.

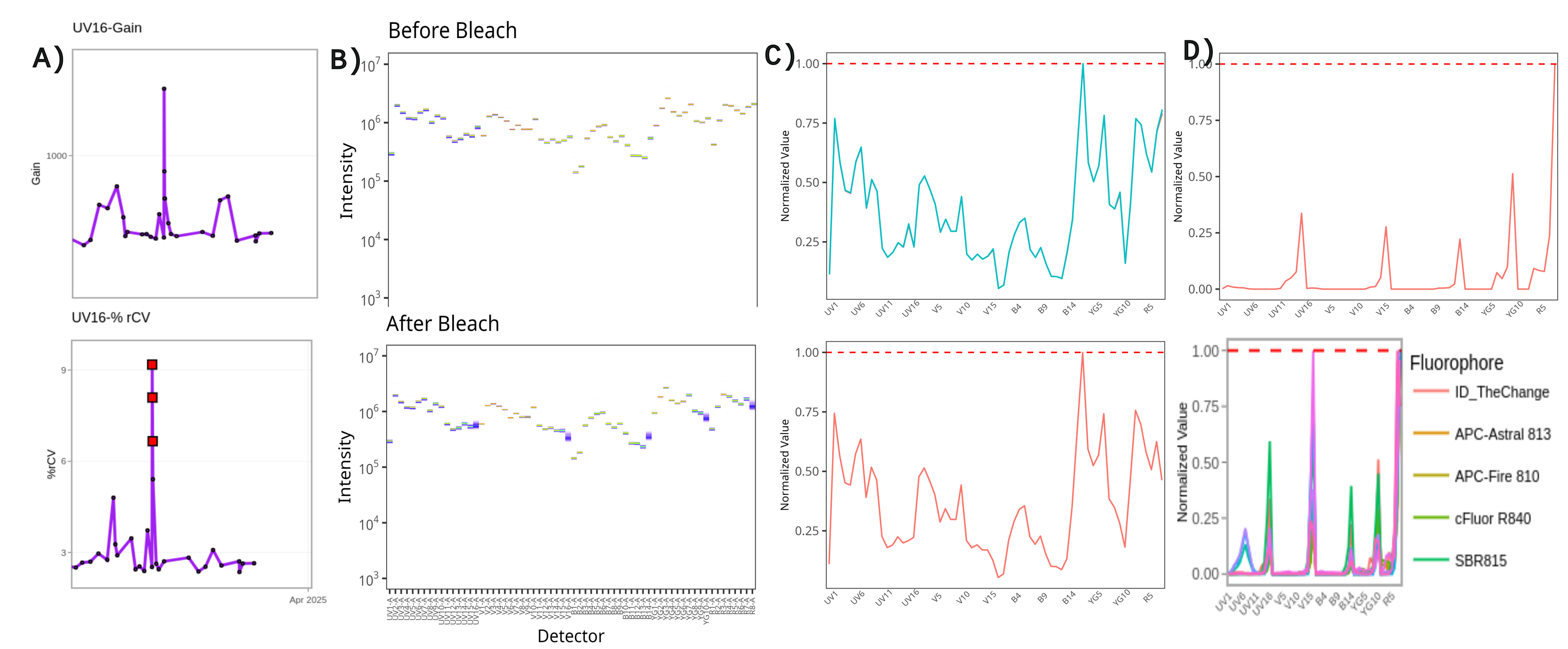
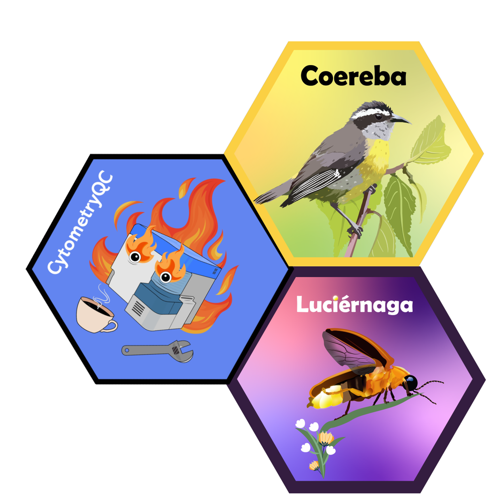

A) UV16-Gain
UV16-% rCV
B) Before Bleach
After Bleach
Intensity
Detector
C)
D)
Normalized Value
Fluorophore
ID_TheChange
APC-Astral 813
APC-Fire 810
cFluor R840
SBR815

Figure 5. Avoiding bleach exposure with SpectroFlo Lot 2006 beads. Following a fluidcs shutdown with bleach, fresh QC beads were run. **A)** Levey-jennings plots show spike in gain and %RCV. In **B)** spectrum plots show bead MFI before (top) and after (bottom) bleach exposure. **C)** lineplots depict grouped normalized signatures from individual cells before and after bleach exposure. In **D)**, the after bleach MFI was subtracted from before bleach bead MFI, and the difference normalized to identifty suspected stripped fluorophore.

Takeaways

Leveraging our access to processed quality control data for each of our instruments, we were able to quantitatively access whether the observed drifts in MFI visible in the Levey-Jennings tracking plots were insignigficant or of concern. We found that drifts in detector MFI do occur by evening for a few detectors unique to each instrument, they are not to the same extent as if the instrument were shut down overnight. When simulating changes to reference controls resulting from these, the changes were minimal, only being visible in a few fluorophores on secondary and tertiary peaks. When we simulated the changes in actual experimental controls and unmixed a full-stained 29-color panel, we did not observe any significant differences. Additionally, using our instruments data we identified when lasers began to fail, and issues with the new QC bead lot so that preventive measures could be taken.

Data availability


Coereba
CytometryQC
Luciérnaga

Would you like to do this your own dataset? The R code used for the analysis and figure generation is available on our GitHub (https://github.com/DavidRach/EveningQC_Cyto2025). The current and historical QC data for all our instruments is available via our website (https://umgccfcss.github.io/InstrumentQC/Data).

Interested in setting up an automated QC tracking website for your own instruments? We have a how-to tutorial for those with some coding literacy (https://github.com/DavidRach/InstrumentQC_Install), and for those without, we are actively finishing an R package that will automatically set one up for Cytek and BD instruments with a single click (https://github.com/DavidRach/CytometryQC).

Take some stickers and track our progress!